Product Optimization & Revenue Contribution Analysis

for Afficionado Coffee Roasters

A Data-Driven Research Paper on Menu Performance, Popularity vs. Profitability, and Revenue Concentration

Prepared for: Unified Mentor

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Tools: Python · Pandas · NumPy · Plotly · Streamlit

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Live Dashboard: analyticsafficionadocoffee.streamlit.app

GitHub Repository

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Abstract

Specialty coffee retailers commonly operate broad menus without a clear, data-driven view of which products actually drive revenue versus which merely drive foot traffic or clutter the menu. This paper presents a product-centric analytics study of Afficionado Coffee Roasters, built on 149,116 point-of-sale transactions spanning three New York store locations (Lower Manhattan, Hell's Kitchen, and Astoria) across 80 SKUs in nine product categories. Using Python (Pandas, NumPy) for computation and Plotly/Streamlit for interactive delivery, the analysis quantifies product popularity, revenue contribution, category dependency, and menu concentration through a Pareto (80/20) lens. Coffee (38.6%) and Tea (28.1%) together account for two-thirds of total revenue of $698,812, while Barista Espresso and Brewed Chai Tea emerge as the two highest-grossing product types. Contrary to a classic 80/20 assumption, revenue is only moderately concentrated: the top 20% of SKUs generate 37.1% of revenue, and roughly 54% of SKUs are required to reach 80% of revenue — indicating a broad-based menu with relatively low single-product dependency risk. The analysis also surfaces "traffic driver" items such as Earl Grey Rg (highest unit volume, 26th in revenue rank) that are popular but not revenue-efficient, and identifies a long tail of Loose Tea and Packaged Chocolate items as candidates for menu simplification. Findings are delivered through an interactive Streamlit dashboard supporting category filters, store selection, and Top-N product ranking, enabling managers to move menu decisions from intuition to evidence.

1. Background and Context

In specialty coffee retail, not all products contribute equally to revenue. Retailers routinely encounter three recurring challenges:

- High-volume products that sell frequently but generate disproportionately low revenue.
- High-priced, premium products that sell infrequently and therefore contribute little in aggregate.
- Overcrowded menus that slow service, confuse customers, and dilute operational focus.

Sustainable retail performance depends on understanding which products customers actually prefer, which products generate the most revenue, and how that revenue is distributed across categories and variants. This project — commissioned for Afficionado Coffee Roasters under Unified Mentor — builds product-centric intelligence to support menu optimization and merchandising strategy.

2. Problem Statement

Although transaction-level product data is available to Afficionado Coffee Roasters, the business lacked structured answers to three questions:

- Clear visibility into product popularity versus profitability.

- Category-level revenue dependency insight.
- Identification of low-impact or underperforming menu items.

Without structured product analytics, menu decisions were largely intuition-driven, creating a risk of operational inefficiency and missed revenue opportunity. This paper addresses that gap directly with a quantitative, reproducible analysis.

3. Objectives

3.1 Primary Objectives

- Identify top-selling and least-selling products.
- Quantify revenue contribution by product and category.
- Measure revenue concentration across the menu (Pareto analysis).

3.2 Secondary Objectives

- Support menu simplification and optimization.
- Identify high-impact "hero" products.
- Highlight low-performing products for review or redesign.

4. Dataset Description

The analysis is built on the transaction-level dataset supplied for the project (Afficionado Coffee Roasters — Transactions), comprising 149,116 rows collected across 2025.

Column	Description
transaction_id	Unique identifier per transaction
year	Transaction year (2025)
transaction_time	Time of transaction (HH:MM:SS)
transaction_qty	Quantity purchased
store_id / store_location	Store identifier and physical location
product_id	Unique identifier for each product
unit_price	Price per unit (USD)
product_category	Broad product group (e.g., Coffee, Tea)
product_type	Product variant within the category
product_detail	Detailed attributes such as flavor, blend, or size

Key dataset characteristics after validation:

- 149,116 transactions across 3 store locations: Lower Manhattan, Hell's Kitchen, and Astoria.
- 9 product categories, 29 product types, and 80 distinct product SKUs (product_id / product_detail combinations).
- No missing values across any column; all transaction quantities and unit prices are positive and within realistic bounds (quantity 1–8 units; unit price $0.80–$45.00).
- Total units sold: 214,470. Total gross revenue: $698,812.33. Average transaction value: $4.69.

5. Analytical Methodology

5.1 Data Ingestion & Validation

The raw CSV was loaded with Pandas and validated for completeness (no nulls), realistic quantity bounds, and consistent product identifiers before any aggregation was performed.

5.2 Revenue Computation

Revenue was computed at the transaction level as:

Revenue = transaction_qty × unit_price

Revenue was then aggregated at three levels — product (SKU), product type, and product category — to support drill-down analysis.

5.3 Product Popularity Analysis

Total units sold were computed per product, and products were ranked independently by sales volume (popularity) and by revenue, enabling direct comparison between the two rankings to surface mismatches.

5.4 Revenue Contribution Analysis

For every product, revenue share (%) of total revenue was computed and compared against its popularity rank to classify items as hero products, premium niche products, traffic drivers, or long-tail products.

5.5 Category & Product-Type Performance

Revenue share by category was computed to assess business dependency on any single category, followed by a breakdown of product-type contribution within each category.

5.6 Revenue Concentration & Menu Balance

A Pareto (80/20) analysis was run on SKU-level revenue to identify revenue anchors versus long-tail products and to evaluate overall menu diversification risk.

6. Key Performance Indicators (KPIs)

KPI	Description
Product Revenue Contribution (%)	Share of total revenue attributable to a product
Product Sales Volume	Popularity indicator (units sold)
Category Revenue Share	Degree of business dependency on a category
Revenue Concentration Ratio	Menu risk exposure (Pareto share)
Product Efficiency Score	Revenue generated per SKU

7. Findings: Category-Level Revenue Contribution

Coffee and Tea dominate the menu, together accounting for 66.7% of total revenue. Coffee alone contributes $269,952 (38.6%) from 89,250 units sold, while Tea contributes $196,406 (28.1%). Bakery and Drinking Chocolate form a secondary tier at roughly 11% each, and five smaller categories — Coffee Beans, Branded, Loose Tea, Flavours, and Packaged Chocolate — together make up only 11.1% of revenue despite representing a meaningful share of the product catalog.

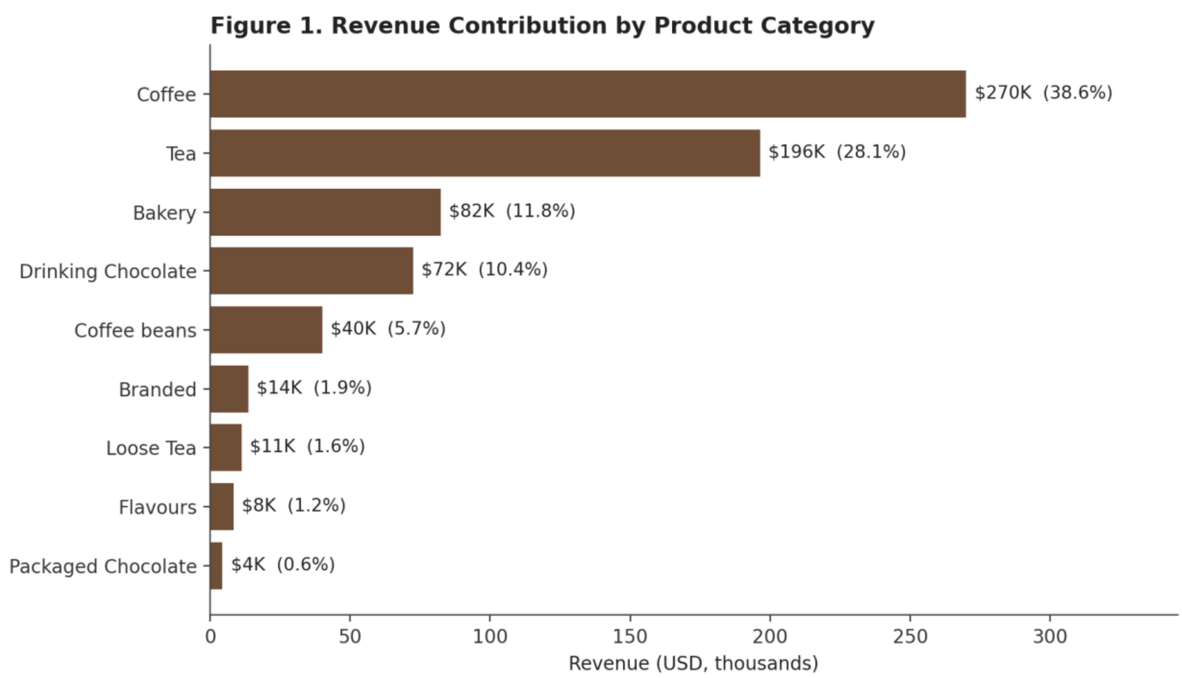

Figure 1. Revenue contribution by product category, ranked by dollar value and percentage share of total revenue.

8. Findings: Top-Performing Products

At the individual product level, Barista Espresso ($91,406) and Brewed Chai Tea ($77,082) are the two largest revenue-generating product types, followed closely by Hot Chocolate ($72,416) and Gourmet Brewed Coffee ($70,035). Among individual SKUs, Sustainably Grown Organic (Large) and Dark Chocolate (Large) — both Drinking Chocolate variants — are the single highest-revenue products, each contributing just over 3% of total revenue on their own.

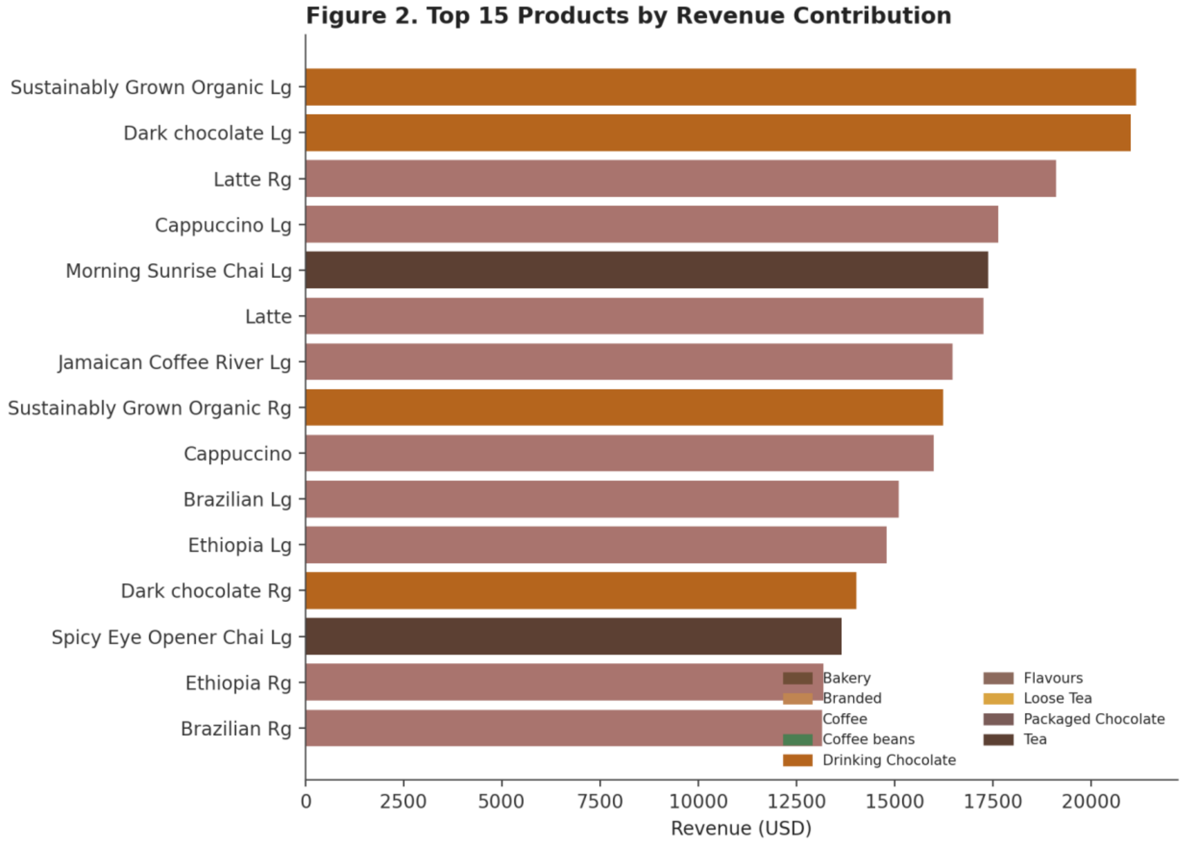

Figure 2. Top 15 individual products (SKUs) ranked by total revenue, colored by product category.

9. Findings: Popularity vs. Revenue — Where the Menu Diverges

Comparing sales volume rank against revenue rank at the product-type level reveals four distinct segments, visualized in Figure 3: hero items that are both high-volume and high-revenue (Barista Espresso, Brewed Chai Tea); premium niche items that sell in low volume but at high value (Premium Beans, Espresso Beans); traffic-driver items that sell in high volume but contribute comparatively little revenue per unit; and long-tail items that are both low-volume and low-revenue.

The clearest example of a volume/revenue mismatch at the individual SKU level is Earl Grey Rg: it is the single most popular product in the entire catalog by units sold (4,708 units, rank #1) yet ranks only 26th by revenue ($11,770). By contrast, Sustainably Grown Organic Lg ranks 11th in units sold but 1st in revenue, reflecting a materially higher price point per unit. This gap illustrates why popularity alone is an unreliable proxy for profitability, and why revenue-per-unit needs to sit alongside volume in any menu review.

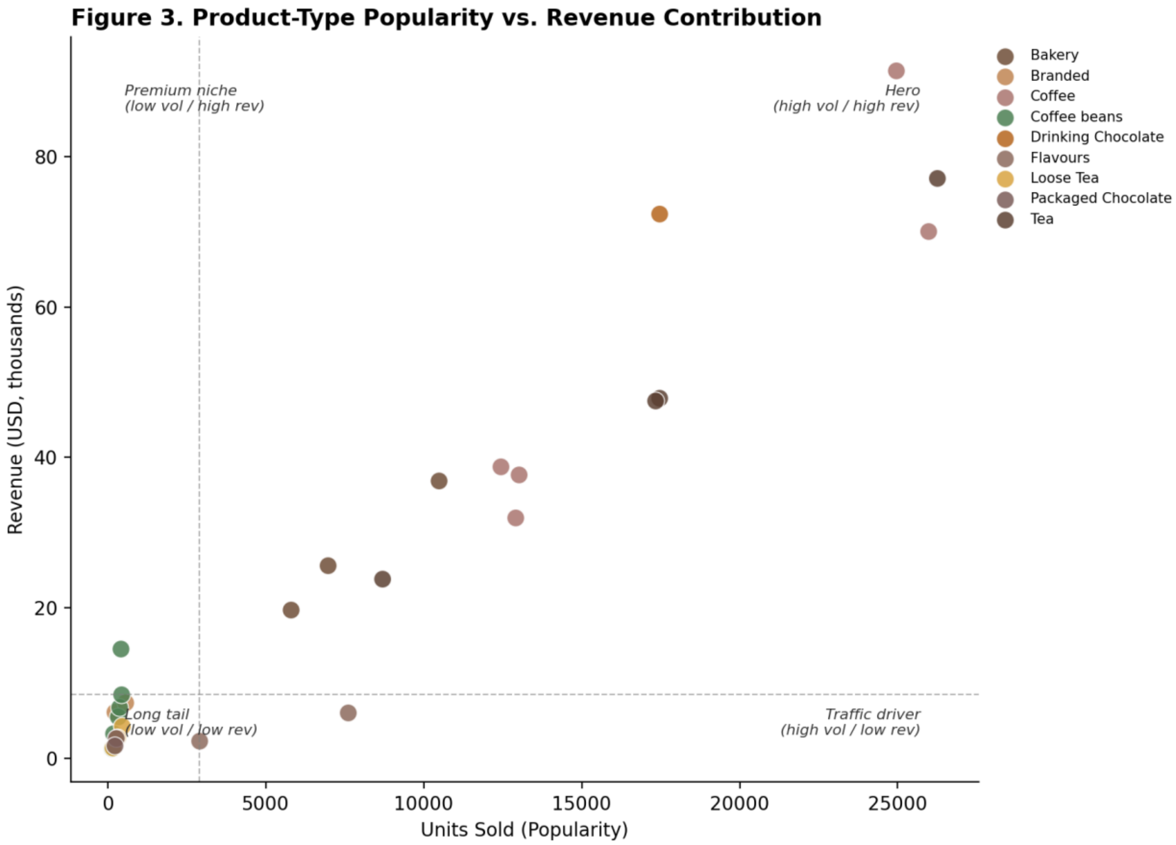

Figure 3. Popularity (units sold) vs. revenue contribution at the product-type level, quadrant-annotated to separate hero, premium-niche, traffic-driver, and long-tail products.

10. Findings: Revenue Concentration (Pareto Analysis)

A classic retail assumption is that roughly 20% of products drive 80% of revenue. The Afficionado Coffee Roasters menu does not strongly follow this pattern: the top 20% of SKUs (16 of 80 products) generate only 37.1% of total revenue, and it takes 54% of SKUs (43 of 80) to reach the 80% revenue threshold. This indicates a comparatively flat, broad-based revenue distribution across the menu rather than dependence on a small number of hero products.

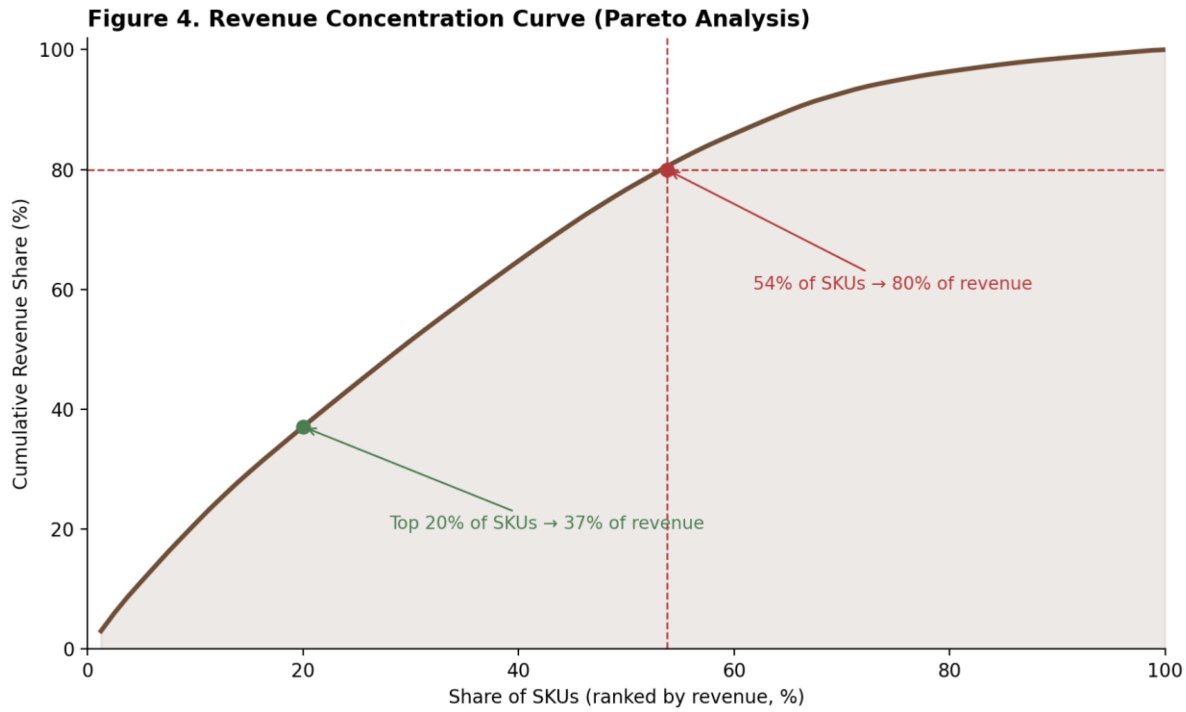

Figure 4. Cumulative revenue share versus share of SKUs (ranked by revenue), showing that the menu's revenue concentration is moderate rather than sharply Pareto-distributed.

From a risk perspective, this is a favorable finding: no single product failure would meaningfully damage total revenue. From an efficiency perspective, however, it also means the business is not benefiting from a small set of highly optimized hero items, and there is room to grow revenue-per-SKU by trimming the bottom tier and reinvesting operational focus into mid-tier performers with strong per-unit economics.

11. Findings: Store Location and Timing Patterns

Revenue is evenly distributed across the three store locations — Hell's Kitchen (33.8%), Astoria (33.2%), and Lower Manhattan (32.9%) — indicating low geographic concentration risk and consistent underlying demand across sites. Lower Manhattan shows the highest average basket value ($4.81) despite having the fewest transactions, suggesting a customer base that purchases higher-value items per visit.

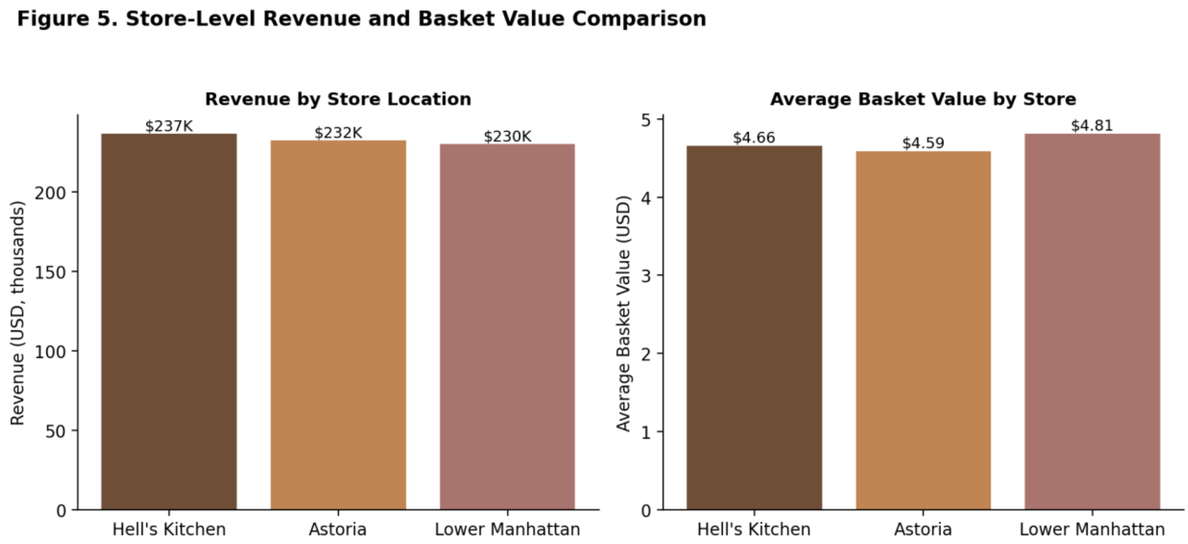

Figure 5. Revenue and average basket value by store location.

Sales volume follows a clear daily rhythm: revenue peaks sharply between 8:00 AM and 10:00 AM (the morning coffee rush), before settling into a steady, lower plateau through the afternoon and tapering off after 7:00 PM.

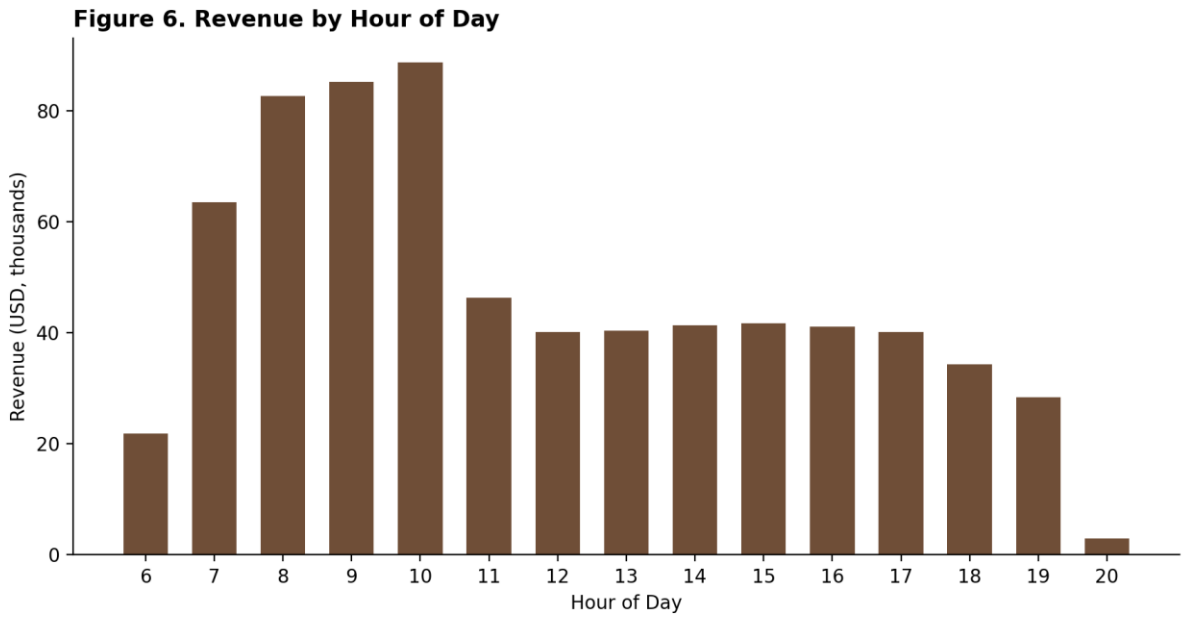

Figure 6. Total revenue by hour of day, showing a pronounced morning peak.

12. The Streamlit Analytics Dashboard

The findings in this paper are operationalized in a live, interactive Streamlit web application that lets store managers explore the same analysis without writing code:

- Product ranking by volume and by revenue, with a Top-N slider to control how many products are displayed.

- Category revenue distribution, with category and product-type filters.
- An interactive popularity-vs-revenue scatter plot mirroring Figure 3, built with Plotly for hover-level product detail.
- A store location selector to compare performance across Lower Manhattan, Hell's Kitchen, and Astoria.
- Product drill-down tables with export functionality for further offline analysis.

Live dashboard: analyticsafficionadocoffee.streamlit.app

Source code: GitHub repository

The application is built entirely on Python (Pandas and NumPy for data preparation, Plotly for charting, Streamlit for the interface), with the same real transaction dataset used throughout this paper as its default data source, and a synthetic-data fallback so the dashboard remains usable if the underlying CSV is unavailable.

13. Business Recommendations

13.1 Menu Simplification

Loose Tea and Packaged Chocolate variants dominate the bottom of the product list by revenue (e.g., Dark Chocolate, Earl Grey, Spicy Eye Opener Chai, Guatemalan Sustainably Grown, Lemon Grass — each under $1,500 in annual revenue). These items are strong candidates for consolidation into fewer SKUs, or removal, to reduce menu complexity and inventory overhead without materially affecting total revenue.

13.2 Re-price or Reposition Traffic-Driver Items

High-volume, low-revenue items such as Earl Grey Rg drive footfall and repeat visits but underperform on a per-unit basis. Consider modest price adjustments, upsell prompts (e.g., pairing with a bakery item), or repositioning as a loyalty/value offering rather than a core revenue line.

13.3 Protect and Promote Hero Products

Barista Espresso, Brewed Chai Tea, Hot Chocolate, and Gourmet Brewed Coffee collectively anchor the menu. These should receive priority placement, consistent quality control, and be safeguarded from supply disruption given their outsized contribution to revenue.

13.4 Leverage Even Store Performance

Because revenue is balanced across all three locations, successful promotions or menu changes piloted at one store are likely to generalize well to the others, reducing the risk associated with store-specific rollouts.

13.5 Staff and Inventory Planning Around Peak Hours

The pronounced 8–10 AM revenue peak suggests staffing and inventory (particularly high-turnover items like espresso and drip coffee) should be weighted toward the morning window to avoid service slowdowns during the busiest period.

14. Conclusion

This project delivers actionable product intelligence by shifting focus from when customers buy to what they buy and what actually drives revenue. Rather than confirming an intuitive 80/20 hero-product pattern, the analysis reveals that Afficionado Coffee Roasters operates a comparatively well-diversified menu, with Coffee and Tea as the anchor categories, balanced performance across all three store locations, and a clearly identifiable long tail of underperforming Loose Tea and Packaged Chocolate items. By combining rigorous EDA with an interactive Streamlit dashboard, the business now has a repeatable, evidence-based framework for ongoing menu optimization, merchandising strategy, and operational planning — replacing intuition-driven decisions with structured, data-backed ones.

15. References & Project Links

- Project brief: Product Optimization & Revenue Contribution Analysis for Afficionado Coffee Roasters (Unified Mentor).

Live Streamlit dashboard: https://analyticsafficionadocoffee.streamlit.app/

GitHub repository (code, notebooks, dataset): github.com/Devangdaksh/Product-Optimization-Revenue-Contribution-Analytics-Dashboard-Python-Pandas-Plotly-Streamlit